

Krishna Subedi

Applied AI engineer — agent platforms & robotics; sole author of 4 arXiv papers

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EXPERIENCE

Neryva

Jul 2025 – Present

Member of Technical Staff · Remote

- Architected the production multi-tenant agent runtime: asynchronous orchestration of agent lifecycles, tool-calling loops, and state-machine transitions with hierarchical memory — LLM output treated as untrusted code, executed in isolated sandboxes to run many agents concurrently per customer.
- Built custom agent harnesses customised to enforce safety constraints, with layered input/output safety (token-classification at ingress flags prompt-injection and jailbreak attempts; output validation at egress) and evaluation pipelines for tool-calling accuracy and context-retrieval drift. Demo: neryva.com.

Smaitic Labs

Apr 2024 – Jul 2025

Full-stack Developer · Kathmandu, Nepal

- Built a three-tier admin hierarchy (main/state/seller) with role-specific access and per-tier dashboards, plus a multi-tenant Spring Boot backend with hierarchical RBAC, ZipCode-based delivery logistics, and a commission engine.
- Shipped [Vanidya](#) marketplace apps: native Android (Kotlin), offline-first PWA, real-time order tracking.

Nepal Oil Corporation Limited

Oct 2023 – Mar 2024

Software Engineer · Kathmandu, Nepal

- Designed a time-series fuel-demand forecasting system for regional fuel demand (diesel, petrol, aviation), modeling festival surges, strikes, and protests so supply stayed smooth and swings never reached the pump.
- Built an automated dispatch reporting system for national fuel logistics, consolidating daily depot and tanker movements into live per-depot dashboards that replaced manual end-of-day report preparation.

PUBLICATIONS

Bounded Precision-Geometry Scaling for Robust Multi-Task Learning

Aug 2026

Sole Author · [arXiv:2608.21653](https://arxiv.org/abs/2608.21653)

- Held macro $0.777 \rightarrow 0.778$ under $1000\times$ loss rescaling, outperforming Nash-MTL on NYUv2.

The Reliability of LLMs for Medical Diagnosis

Mar 2025

Sole Author · [arXiv:2503.10647](https://arxiv.org/abs/2503.10647)

- Benchmarked frontier LLMs across 52 clinical scenarios; quantified adversarial prompt vulnerability.

PROJECTS

Agent Studio: Enterprise AI Agent Platform

Jul 2025 – Present

Work Project & Open Source · ML & Platform

- Architected a multi-tenant LLM control plane on the CoALA framework — tenant memory, action space, and LangGraph decision loop — with per-tenant vector-isolated RAG and an LLM gateway handling provider routing, fallback chains, circuit breakers, and cost ledgers for customer-selected models.
- Built the session/context engine (durable threads, compaction, fork) and eval stack (Garak scans, RAGAS suites) wrapped in Presidio PII redaction and NeMo Guardrails validation with low-confidence human escalation.

Krishi Vaidya: AI-Powered Crop Disease Diagnosis

Apr – Jul 2026

Personal Project · ML

- Built a dual-model crop-disease diagnosis system: LoRA/PEFT fine-tuned Qwen2.5-VL-3B for structured diagnosis and a YOLO-family real-time detector deployed on Lightning.ai.
- Optimized inference for edge and cloud: compressed the detector to INT8 TFLite for on-device OpenCV inference; served the VLM on GPU-aware vLLM with 4-bit NF4 quantization and Flash Attention 2, enforcing structured JSON with confidence and out-of-distribution detection.

EDUCATION

Bachelor of Engineering in Mechanical Engineering

2019 – 2023

Tribhuvan University · Kathmandu, Nepal · Licensed PE

SKILLS

Languages: Python, C/C++, SQL · **AI/ML:** PyTorch, LLMs/GenAI, Time-Series, Multi-Task & Continual Learning, LLM Evaluation · **Agents:** LangGraph, FastAPI, RAG, Presidio, NeMo Guardrails, Langfuse, Multi-Tenancy · **Robotics:** ROS, PyBullet, Unity, Inverse Kinematics, Motion Planning · **DevOps:** Docker, Kubernetes, Git, CI/CD, Linux, Redis, PostgreSQL